

RESEARCH

Soybean Irrigation Management: Agronomic Impacts of Deferred, Deficit, and Full-Season Strategies

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ABSTRACT

The efficiency of irrigation water use is improved if irrigations are scheduled only when soil water depletion exceeds a chosen crop-specific threshold percentage of field capacity (FC). Further improvement may be possible by accounting for seasonal variance in yield sensitivity to water stress. In soybean [*Glycine max* (L.) Merr.], the vegetative and bloom stages are notably less sensitive to water stress than are the pod-set and seed-fill stages. A 2-yr experiment was conducted to determine the yield impact of deferring irrigation until the beginning-pod stage (R3). Eight soybean cultivars were evaluated in subplots within four main plot irrigation treatments: (i) full-season control, with a 35% FC soil-water depletion trigger to schedule irrigation that replenished 100% of the crop evapotranspiration (ET) loss since the prior irrigation; (ii) a defer-irrigation-to-R3 strategy with the 35% FC depletion trigger ignored until the onset of stage R3; (iii) a deficit irrigation strategy with irrigation events on the same dates as those in the full-season control, but limited to two-thirds of the control amount; and (iv) rainfed control. Below-normal spring rainfall in each year resulted in soil water depletion greater than 35% FC before irrigation commenced in the R3-deferred strategy but did not impact yield versus the full-season control yield. A season-long deficit irrigation strategy significantly reduced yield below the full-season control, though it still produced significantly more yield than the rainfed control. These findings suggest that differentials in sensitivity of the crop developmental stage should be incorporated into irrigation scheduling models.

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Abbreviations: C, cultivar; DAE, days after emergence; DF, deficit; ET, evapotranspiration; FC, field capacity; FS, full season; I, irrigation; IN, indeterminate cultivar; IWUE, irrigation water-use efficiency; MG, maturity group; NI, nonirrigated; PAW, plant available water; PWP, permanent wilting point; SD, semideterminate cultivar.

IRRIGATION OF SOYBEAN [*Glycine max* (L.) Merr.] in the Northern Great Plains of the USA occurs primarily in Nebraska, Kansas, South Dakota, North Dakota, and portions of the neighboring states. In Nebraska, about 50% of the state's total soybean area (2.02 million ha) was irrigated during the past decade. In the three other states, less than 15% of the total harvested soybean hectareage was irrigated (<http://quickstats.nass.usda.gov/>).

The seasonal cumulative evapotranspiration (ET) pattern for a soybean crop in a given field will depend on (i) the number of days between the date of crop emergence and date of physiological maturity (stage R7), and the latter date will depend on the cultivar maturity group (MG) chosen; (ii) the degree to which early spring snowmelt and preemergence rainfall have restored the soil water content to 100% FC in the 90- to 120-cm soil depth that will be accessed by soybean root hairs during the growing season; and (iii) the total amount of rainfall received at the given

Published in Crop Sci. 54:2782–2795 (2014).

doi: 10.2135/cropsci2014.03.0261

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field during the period from crop emergence to physiological maturity in the given crop year. Although the total amount of seasonal rainfall may be near normal for that crop-development time frame, it may be ill-distributed in terms of days between successive rainfall events, which can result in yield-impacting crop water stress on those days when sufficient soil water is not available for daily crop transpiration. Applying no more irrigation than required on a timely basis to meet this transpiration demand would be consistent with the now well-established linear relationship between soybean yield and the amount of water supplied for seasonal transpiration, which was first documented in soybean by Specht et al. (1986, 2001).

Historically, irrigation scheduling protocols had to be implemented via manual calculations (e.g., Lundstrom and Stegman, 1988; Wright, 2002; Melvin and Yonts, 2009), but with the advent of computers, spreadsheet software designed to do those scheduling calculations became available for installation on producer and crop-consultant computers, although manual input of daily weather data was still needed (e.g., Steele et al., 2010; Joern and Hess, 2013). More recently, producer-accessible websites (e.g., <http://www.hprcc3.unl.edu/soywater>) have become available that provide free virtual access to daily weather data from a nearby automated weather station. In any case, the primary objective of irrigation scheduling is to maintain a soil water content above the midpoint between 100% FC, which is the maximum amount of water a soil can hold against gravity, and 0% FC, known as the permanent wilting point (PWP), which occurs when the soil water content is so dry that plant root hairs cannot acquire soil water, leading to plant death (Hillel, 1998; see Fig. 1 in this website: [http://www1.agric.gov.ab.ca/\\$department/deptdocs.nsf/all/ind10761/\\$file/enhance_quality_and_yield.pdf?](http://www1.agric.gov.ab.ca/$department/deptdocs.nsf/all/ind10761/$file/enhance_quality_and_yield.pdf?)). The soil water content difference between those two extremes is termed plant-available water (PAW). Irrigation events are scheduled (i.e., triggered) on any date when the cumulative amount of daily crop ET exceeds a producer-chosen percentage of allowable PAW depletion in the crop root zone (see Table 1 in Irmak et al., 2014a). Using a trigger threshold of greater than 50% FC depletion is not recommended for commencing irrigation, except possibly in deficit irrigation strategies (Payero et al., 2005), not only because PAW becomes appreciably less available to plant root hairs as the FC percentage approaches its zero (PWP) limit, but also because it may take several days to make a full-circle application with a center pivot system, and by the time the pivot completes the circle, the PAW in some parts of the field may have been depleted by more than 50%, causing plant stress. Although an irrigation trigger of 35% FC depletion may seem to be an inefficient choice because it leads to more frequent irrigation events, that choice may be necessary when (i) the irrigation system capacity imposes a limit on the amount

of water deliverable per irrigation event, or (ii) the field soil possesses low water-holding capacity (e.g., a sandy soil), wherein just a few days of daily crop soil-water extraction can result in a quick return to a 50% FC level very soon after a rainfall or irrigation event.

It is now well known that irrigation scheduling based on monitoring of the soil water balance greatly improves irrigation water-use efficiency (IWUE), which is typically defined by the ratio of irrigated minus rainfed seed yield to the total seasonal amount of irrigation applied. Yet, further IWUE improvement is still possible if the sensitivity of the crop development stages to yield reduction with coincidently timed water stress is considered. Instead of keeping the irrigation-triggering FC depletion percentage constant, it might be possible to seasonally adjust it to be more congruent with the water-stress sensitivity of the crop during its seasonal development. Van Doren and Reicosky (1987) were the first to note the benefit of this approach for soybean, with a later concurrence by Heatherly and Elmore (2004), that “sensitivity of the plant to water-deficit stress should signal when to irrigate, with soil water status determining how much to irrigate.”

Irrigation-timing research conducted with furrow and sprinkler systems in the Great Plains region has documented a differential yield responsiveness of soybean (and other crops) to irrigation or water-stress events when these events are timed to have a coincident occurrence with successive reproductive developmental stages (e.g., Korte et al., 1983a, 1983b; Kadhem et al., 1985a, 1985b; Muchow et al., 1986; Randall and Sinclair, 1988; Klocke et al., 1989; Specht et al., 1989; Andriani et al., 1991; Sadras and Milroy, 1996; Andresen et al., 2001; Irmak et al., 2014b). Surprisingly, soybean irrigation events preceding or coinciding with flowering, particularly at stage R2 (full bloom), did not enhance seed yield, and in fact, in some cultivars a yield penalty was observed. This finding implied that irrigation could potentially be deferred until after R2 without much impact of this deferral on seed yield, so long as irrigation was subsequently commenced at stage R3 (beginning pod) and continued thereafter. If the yield is not changed but the irrigation amount is reduced, which is possible if pre-R3 irrigation is avoided without a yield reduction, then the IWUE is improved.

In the present study, the primary goal was to precisely quantitate the degree to which the yield might be reduced if the *first* irrigation of the season was deferred until the soybean crop attained stage R3 but with subsequent irrigation events appropriately scheduled thereafter to replace ET. Prior studies employed furrow or sprinkler irrigation, but drip irrigation was used here because it offers greater precision in delivering a given water amount and requires less field space for an irrigation experiment, both of which help minimize experimental error in the data analyses.

MATERIALS AND METHODS

This experiment was conducted in 2006 and 2007 in an Agronomy Farm field located on the East Campus of the University of Nebraska at Lincoln, NE (40°50'12" N, 96°39'44" W). The soil at the field site is classified as a deep fine-silty, mixed, superactive, mesic Cumulic Hapludoll Kennebec silt loam (<http://websoilsurvey.sc.egov.usda.gov/App/HomePage.htm>). Maize (*Zea mays* L.) was the prior crop each year. The field was fall-plowed after the maize harvest then field-cultivated twice in the spring of each year. Before planting on 27 Apr. 2006 and 30 Apr. 2007, the soybean seed was treated with the fungicides fludioxonil (4-(2,2-difluoro-1,3-benzodioxol-4-yl)-1*H*-pyrrole-3-carbonitrile) and mefenoxam ((*R,S*)-2-[(2,6-dimethylphenyl)-methoxyacetyl-amino]-propionic acid methyl ester) and the insecticide thiomethoxam ((*EZ*)-3-(2-chloro-1,3-thiazol-5-ylmethyl)-5-methyl-1,3,5-oxadiazinan-4-ylidene(nitro)amine). For weed control, the field was preplant treated with 0.03 kg ha⁻¹ of flumetsulam (*N*-[2,6-difluorophenyl]-5-methyl-[1,2,4]-triazolo-[1,5*a*]-pyrimidine-2-sulfonamide) and 1.07 kg ha⁻¹ of *S*-metolachlor (2-chloro-*N*-[2-methylphenyl]-*N*-[(1*S*)-2-methoxy-1-methylethyl] acetamide). A few broadleaf weed escapes were rogued after soybean emergence on 9 May 2006 and 8 May 2007.

The experimental design each year was a randomized complete block design with four replicates. The treatment design was a split plot, with four water regimes as main plots and eight soybean cultivars as subplots. The experimental units were 4.3-m long six-row subplots, with an interrow spacing of 0.76 m. Four cultivars of about MG 2.6 to 2.7 (i.e., three indeterminates: Asgrow AG2703, Dekalb DKB26-52, and Pioneer P92M61; and one semideterminate: Nebraska U01-390489) and four of about MG 3.0 to 3.1 (i.e., three indeterminates: Asgrow AG3101, Dekalb DKB31-51, and Pioneer P93M11; and one semideterminate: Nebraska NE3001) were randomly assigned to the eight subplots in each of the four main plots, which, in turn, were randomly assigned to each replicate. The seeding rate was set to 340,000 viable seeds ha⁻¹ but adjusted upward each year for each cultivar based on its germination percentage (which varied from 80 to 90%). The four water regimes were (i) a full-season (FS) irrigation strategy (designated 100ET-FS), in which seasonal irrigation events were triggered on any calendar date when the crop soil-water depletion in the 0.9-m root zone exceeded 35% FC soil water content, with the water application amount at each event set to be commensurate with the amount needed to replace about 100% of the crop ET amount accumulated since a prior rainfall or irrigation event; (ii) a deferred irrigation strategy (100ET-R3), in which the first seasonal irrigation was intentionally not applied until soybean stage R3—which was attained in early July each year—but with irrigation events thereafter scheduled just like those of the 100ET-FS control (i.e., 35% FC depletion trigger and 100% ET replacement basis); (iii) a deficit irrigation strategy (066ET-DF), in which irrigation events coincided in timing with those in the 100ET-FS strategy but with the water application amount applied at each event limited to just two-thirds of the 100ET-FS applied amount; and (iv) a nonirrigated (NI) treatment (000ET-NI), which provided a baseline estimate of the yield achievable in a rainfed-only water regime. The defer-to-R3 irrigation strategy (referred to more succinctly as just the R3

strategy from here on) was thus ultimately compared with the two control treatments—no irrigation, and a full-season irrigation treatment, in which irrigation events were scheduled without regard to the crop developmental stage coincident with the first irrigation event. Although the R3 strategy is an irrigation deferral strategy, it can result in a soil water deficit before irrigation is applied at R3, so comparing it with a deficit irrigation strategy was appropriate.

The six-row-wide, eight-cultivar subplots were arrayed end to end within a main plot, thereby permitting irrigation of the main plots to be accomplished with drip-tape lines placed on the soil surface adjacent to plants in all rows of the six-row main plot. The drip-tape line emitter-to-emitter spacing interval was 30 cm. The rate of water application was 0.267 cm ha⁻¹ h⁻¹.

The daily crop ET was computed to estimate the daily soybean soil water depletion from stages VE to R7 using the Penman equation (<http://www.hprcc.unl.edu/awdn/et/>) and daily inputs of solar radiation, temperature, humidity, and wind speed at an automated weather station (http://www.hprcc.unl.edu/stations/index.php?action=metadata&network_station_id=a254809) located within 400 m of the experimental field site. The daily crop ET was successively summed to obtain the cumulative daily crop ET. Recorded rainfall amounts at the nearby weather station were validated with an in-field rain gauge.

The maximum soil-water-holding capacity (between 100% and 0% FC, PAW) for the soil texture type in this field was about 5.1 cm per 0.30 m of soil depth. The stored soil-water content at planting and emergence in each season was at or near 100% FC, due to spring snowmelt and preplant rainfall. Rain deposition rates during rainfall events did not exceed the 50 mm h⁻¹ water infiltration rate for this soil, and rainfall run-off from this zero-slope field site was not observed.

Reproductive stages were tracked in each cultivar on a biweekly basis using the staging system first described by Fehr et al. (1971) and later graphically by Fehr and Caviness (1977). Irrigation could not be applied separately to each of the eight cultivar subplots, but because the date of attainment of R3 did not vary by more than 5 d between cultivars, the date of R3 for P93M11 was arbitrarily chosen for commencing the first irrigation in each replicated 100ET-R3 treatment main plot containing all eight cultivar subplots. The last irrigation event (and amount) scheduled each year was based on projecting a probable R7 date for P93M11, using the method of Torrión and Specht (2011), for estimating the water amount needed in last seasonal irrigation event to allow the crop to draw down the soil moisture to 50% FC on the day the crop was expected to attain R7 physiological maturity (Fehr et al., 1971).

Seed yield and other trait data were collected from the central two rows of each of the six-row subplots after the cultivars attained an R8 date of full maturity (when 95% of pods in the two central rows were judged mature). Maturity was expressed as days from planting to R8 maturity. Mature plant height (cm) was measured as the distance from the ground surface to the main stem apex of several plants in the two center rows. The mature plant population was based on a mature plant count taken in a 4.3-m section of a one-subplot row. Lodging was visually judged with a scoring system ranging from 1 (all plants erect) to 5 (all plants prostrate). Seed yield (kg ha⁻¹) on a 13%

seed moisture basis was estimated from the seed weight and moisture electronically recorded by the plot-combine on harvest of each subplot. The mean weight (g) of two independent 100-random-seed samples, adjusted to 13% moisture, was used as a measure of seed mass. Protein and oil contents in a 75-g seed sample, adjusted to 0% moisture, were obtained for each subplot by a near-infrared analytic method (Hymowitz et al., 1974).

For statistical analysis of this split-plot randomized complete block design, SAS PROC MIXED METHOD = TYPE3 (Littell et al., 1996; SAS Institute, 2008) was used with a MODEL statement containing the ALPHA = 0.05 and DDFM = SATTERTHWAIT options. Block (i.e., replicate) was treated as a RANDOM effect, but year (Y), irrigation (I), and cultivar (C) were treated as fixed effects. An ESTIMATE statement was used to derive means for the main effects (I and C) and I $\times$ C interaction and also to compute the difference between effects specified in preplanned single-degree-of-freedom CONTRAST statements that were designed to evaluate the cultivar source of variation in terms of the impact on each trait of (i) early- versus late-maturing cultivars, and (ii) indeterminate versus semideterminate cultivars. The LSMEANS statement included the ADJUST = TUKEY option for using the Tukey RangeTest (adjusted for multiple comparisons) to evaluate a ranked (high to low) ordering of main effect means for significant differences, coupled with the SLICE option for assessing the significance of I or C simple effects in the I $\times$ C interaction term. Unfortunately, when collecting maturity date data in 2007, we discovered that, due to human error, the cultivar P93M61 (MG 3.6) had been inadvertently planted that year, instead of the intended cultivar P92M61 (MG 2.6), which was planted in 2006. Interpretability of single-year data analyses was not impacted by this error, but the data for these two cultivars, plus the data for the comparable P93M11 (MG 3.1) cultivar, had to be dropped to conduct a (cultivar) maturity-balanced, combined 2-yr data analysis and the two preplanned PROC MIXED contrasts therein.

RESULTS AND DISCUSSION

Seasonal Weather

Monthly and seasonal means for the weather variables of air temperature, rainfall, and ET for the test-year seasons of 2006 and 2007 are provided in Table 1 for comparison with historical 27-yr means (mid-1986 to 2013) at this weather station. The air temperature means for June, July, August, and the full season were not markedly different between the 2 yr and also were not much different from historical means. In 2007, except for a warmer-than-normal May, the monthly temperature means were near normal; however, May and September were notably colder than normal in 2006.

Total season rainfall was below normal in 2006 (Table 1), primarily because of low rainfall in May, June, and July, whereas in 2007, the total season rainfall was above normal, but only because August rainfall in 2007 was nearly double the normal amount. The historical May and June rainfall amounts totaled 19.9 cm, but they totaled

Table 1. Monthly and seasonal means for air temperature, total rainfall, and total alfalfa-referenced evapotranspiration (ET) for 2006 and 2007.[†]

Variable and year	May	June	July	Aug.	Sept.	Season
Mean air temperature	°C					
2006	10.3	23.9	27.1	24.4	17.3	22.2
2007	19.0	22.5	26.1	24.2	19.6	22.7
1986–2012	17.5	23.0	25.6	24.2	19.2	21.9
Total precipitation	cm					
2006	3.8	3.3	6.8	10.6	10.7	35.2
2007	10.9	2.8	3.7	15.4	8.1	48.9
1986–2012	10.2	9.7	8.3	8.2	7.1	43.5
Total ET	cm					
2006	18.2	20.3	21.2	14.5	11.0	85.7
2007	15.7	17.7	19.2	15.1	13.0	80.7
1986–2012	16.5	18.5	19.0	16.4	13.1	83.4

[†] Data recorded at an automated weather station located about 400 m distant from the experimental site. The station's historic 26-yr means for these three variables are also presented. See Figures 1A and 1B for the graphs of daily precipitation and daily soybean-specific ET values.

only 7.1 cm in 2006 and 13.7 cm in 2007. The occurrence of below-normal rainfall in the two test years (respectively 36 and 69% less) from the late April planting dates to early July R3 dates was unexpected, but resulted in a May–June rainfall deficit that was fortuitously suitable for rigorously evaluating the yield impact of deferring irrigation until the crop attained R3 of July 5 in each year.

Total seasonal ET was higher than normal in 2006 (Table 1), primarily because of high ET values in May, June, and July. In contrast, the total seasonal ET was lower than normal in 2007, due to low ET values in May, June, and August of that year.

Crop Water Use

The seasonal curves for cumulative daily crop water use (i.e., Penman-equation-estimated soybean ET) and for the cumulative amount of applied irrigation (plus rainfall) in the 100ET-FS, 100ET-R3, 066ET-DF, and 000ET-NI treatments, are presented respectively as brown, green, blue, red, and black lines in Fig. 1A (2006) and Fig. 1B (2007). The corresponding green, blue, and red bars in Fig. 1C and Fig. 1D denote the irrigation dates and amounts for the three irrigation treatments, whereas the black bars denote rainfall dates and amounts. Single-irrigation events were applied overnight, so for technical correctness, the before- and after-midnight water amounts for these single events are shown in Fig. 1C and 1D as two separate bars on successive days. On several occasions each year, soil water contents greater than 100% FC were observed when a rainfall event occurred soon after a prior rainfall or irrigation event. The test-site soil accommodated some additional rain water because its saturation capacity above 100% FC is about 6.5 cm per 0.3 m of soil depth. Of course, soil water in excess of 100% FC rapidly

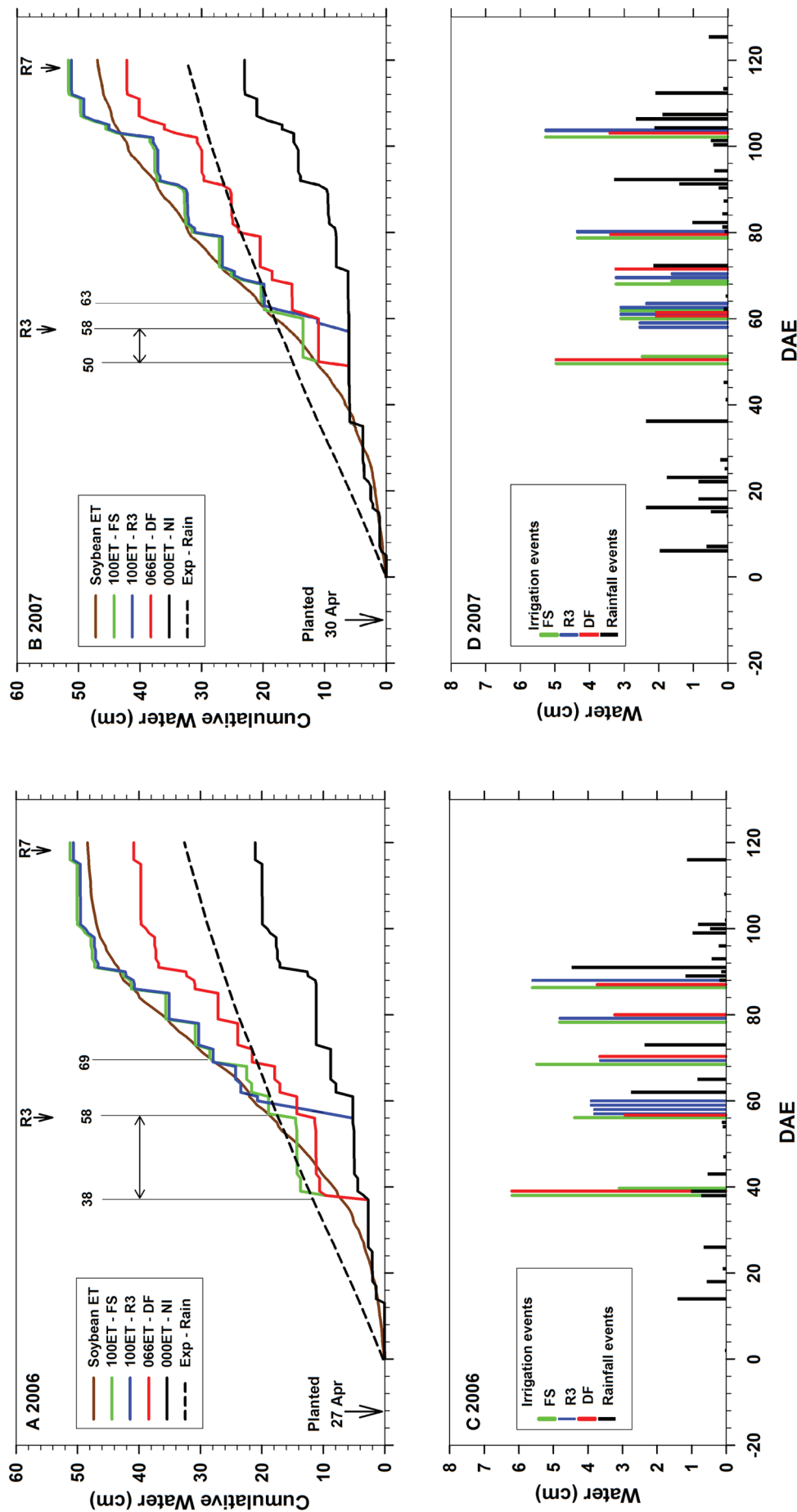


Figure 1. Graph of 2006 (A) and 2007 (B) cumulative daily crop soybean evapotranspiration (ET; brown line) and cumulative seasonal water supply (irrigation plus rainfall amount) for ET-based irrigation scheduling treatments of full-season (FS) irrigation, 100ET-FS (green line); irrigation deferred to the R3 beginning pod stage, 100ET-R3 (blue line); deficit irrigation (DF), 066ET-DF (red line); and non-irrigation (NI). 000ET-NI (black line) versus days after emergence (DAE), from DAE = 0 to DAE = about 120 (R7 physiological maturity). The dashed straight line indicates the cumulative average rainfall that can be expected on a DAE basis, based on historical data from the LINCOLNIAR weather station (ID# a254809) managed by the High Plains Regional Climate Center. The first two adjacent vertical lines, connected with a connecting horizontal double arrowed line denote the DAE of the first applied irrigation event in 100ET-FS, and the DAE of stage R3 in both the 100ET-FS and 100ET-R3 treatments, whereas the third vertical line denotes the DAE at which the cumulative seasonal water supply for those two treatments attained equivalence. (C) Bar chart of 2006 and (D) bar chart of 2007 (B) rainfall and irrigation dates and amounts. Bar color corresponds with the color of curves in (A) and (B).

drains downward in response to gravity within about 48 h, ultimately returning the soil water content to just the amount that can be held against gravity (i.e., 100% FC). Still, plant root hairs can utilize the excess water to supply crop transpiration during that drainage time. So after an allowance for a 2-d transpiration amount supplied by that excess water during the 48-hr drainage period, the remaining portion of the rainfall amount exceeding 100% FC soil water content was excluded from the green, blue, red, and black seasonal cumulative water supply curves in Fig. 1 A and 1B; thus, only “effective” rainfall is included in these curves. However, the bars in the graphs in Fig. 1C and 1D do show the actual total rainfall amounts.

In the 2006 and 2007 Fig. 1 A and B, the seasonal pattern of cumulative daily (effective) rainfall (i.e., solid black curve) in each year was always less than the historical expectation of cumulative daily rainfall at this site (dashed black straight line). Thus, in both years, the historical total rainfall amount from days after emergence (DAE) = 0 (9 May 2006; 8 May 2007) to the R7 stage DAE = 119 (3 Sept. 2006; 4 Sept. 2007) was about 32–33 cm (i.e., the end of dashed straight line at R7); however, the effective rainfall amount in the same 120-d period in both years was only 21–23 cm (i.e., the end of solid black curve at R7).

Comparison of the seasonal cumulative daily rainfall (black) curve with the seasonal cumulative daily soybean ET (brown) curve indicates that the ET-rain deficit became larger over the course of each growing season, resulting in plant water stress in the rainfed check treatment. Irrigation was scheduled in the 100ET-FS control treatment to provide water (using a 35% FC depletion trigger) to ensure that the cumulative water supply in this treatment (blue curve) kept up with cumulative soybean ET (brown curve). The first irrigation event in FS was auto-triggered on DAE 38 in 2006 and on DAE 50 in 2007. In contrast, the first irrigation in the 100ET-R3 treatment was intentionally deferred until the R3 stage, which occurred on DAE 58 in both years. After that initial irrigation event, irrigations were repeatedly scheduled thereafter, on a 35% FC depletion trigger basis, to ensure that the cumulative water supply in this R3 treatment (green curve) was quickly returned to a condition of keeping up with the soybean ET line (brown curve). To mitigate confounding treatment effects, the FS and R3 treatments were brought to soil moisture equivalence as soon as possible after R3 (DAE = 58 each year). This equivalency was attained on DAE 63 in 2007, 5 d after R3 (Fig. 1B). In 2006, complete equivalency was not precisely attained until DAE 69 in that year (due to an irrigation timer malfunction), but *near* equivalency was attained on DAE 64 (Fig. 1A). Thus, in both years the 100-ET R3 treatment soil moisture at stage R3 was effectively returned within 5 d to the same soil moisture content in the 100-ET FS treatment. Ideally, an unconfounded comparison of final seed yields attained

in these two irrigation strategies would entail achieving a seasonal water-supply equivalency in both strategies immediately on the day at which stage R3 is attained, but in practice, such immediate equivalency is simply not feasible in a field experiment.

At this experimental site, the linear historical trend in cumulative daily rainfall crosses the sigmoid (Penman-equation-derived) pattern of cumulative daily soybean ET just about the time a late-April-planted crop attains R3 (Fig. 1A and 1B). Thus, the cumulative daily rainfall amount normally exceeds the cumulative daily soybean ET during the accelerating phase of the sigmoid soybean ET curve but is then expected to be deficient during the decelerating phase. Between rainfall events, of course, the soybean crop must rely on water stored in those soil zones accessible by root hairs. The amount of this stored water supply can be substantive when the field soil texture has a high water-holding capacity (i.e., silt loam or silty clay loam soil) and has been recharged to 100% FC at the date of planting or emergence, which was the case in 2006 and 2007 in this study and also in most years in the western Corn Belt (Grassini et al., 2010). This is why a defer-to-R3 irrigation strategy was considered worthy of evaluation.

Water Regime Main Effect

The results for single-year analyses of variance are presented in Table 2. Means for the four water regimes are shown in Fig. 2 for just the four traits wherein the main effect was consistently significant ($P < 0.05$) in each year. The interaction of this main effect with cultivars will be discussed in a later section. Water stress hastens senescence, as was previously documented by Specht et al. (1986, 2001), so the earlier maturity in the rainfed nonirrigated (NI) and in the deficit irrigation (DF) treatments was not unexpected (Fig. 2A and 2B).

When applied during vegetative development, irrigation increases the internode length of the main stem in indeterminate cultivars, but not as much in determinate cultivars, because vegetative development continues during the reproductive stages of R1 (beginning bloom) to R3 and then to R5 in the former but ceases at R1 in the latter (Bastidas et al., 2008). Each year, it was consistently observed that the FS treatment had the tallest plants, followed in decreasing order by the R3, DF, and NI treatments (Fig. 2C and 2D), with the NI treatment exhibiting a statistically significant shorter plant height than the three other treatments. Stem-height measurements were more precisely executed in 2007 than in 2006 (i.e., SE_{diff} of 0.12 versus 2.7), leading to a stem-height difference between R3 and DF that was significant in 2007 but not in 2006.

Plant lodging was significantly greater with the FS treatment (Table 2, Fig. 2E and 2F), because irrigation applied in late June led to a lengthening of the internodes in the upper stem, resulting in tall plants that were more

Table 2. Source of variation, numerator (n) df and denominator (d) df for the F-tests and probability values obtained in a PROC MIXED analysis of variance of agronomic traits for irrigation and cultivar main effects and their interaction for each year.[†]

Source of variation	n df	d df	Days to maturity	Stem height	Lodging score	Plant density	Seed protein	Seed oil	Seed mass	Seed yield
2006										
Irrigation (I)	3	9	0.004	<0.001	<0.001	0.501	0.032	0.195	<0.001	<0.001
Cultivar (C)	7	84	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
C: 4E vs. 4L	1	84	<0.001	0.189	<0.001	0.095	<0.001	<0.001	0.001	<0.001
C: 2S vs. 6I	1	84	<0.001	<0.001	<0.001	<0.001	<0.001	0.112	<0.001	0.253
I × C	21	84	0.220	0.031	0.004	0.942	<0.001	<0.001	<0.001	<0.001
2007										
I	3	9	<0.001	<0.001	<0.001	0.242	0.363	<0.001	0.165	<0.001
C	7	84	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
C: 3E vs. 3L	1	84	<0.001	0.811	0.918	<0.001	<0.001	<0.001	<0.001	<0.001
C: 2S vs. 4I	1	84	<0.001	<0.001	<0.001	<0.001	<0.001	0.001	<0.001	<0.001
I × C	21	84	<0.001	<0.001	<0.001	0.891	0.528	<0.001	<0.001	<0.001
2006–2007[‡]										
Year (Y)	1	6	0.011	0.024	0.006	<0.001	0.033	<0.001	<0.001	0.453
I	3	18	<0.001	<0.001	<0.001	0.818	0.079	<0.001	<0.001	<0.001
Y × I	3	18	0.011	0.543	<0.001	0.106	0.241	<0.001	0.004	0.019
C	5	120	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
C: 3E vs. 3L	1	120	<0.001	0.763	0.798	<0.001	<0.001	<0.001	<0.001	<0.001
C: 2S vs. 4I	1	120	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.005
Y × C	5	120	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.141	0.001
I × C	15	120	0.003	<0.001	<0.001	0.909	0.133	<0.001	<0.001	<0.001
Y × I × C	15	120	0.016	0.110	0.012	0.895	0.006	0.126	0.017	0.402

[†] A combined 2-yr analysis is also shown. Single df contrasts between the four (or three) early (E) maturing *versus* four (or three) late (L) maturing cultivar brand name pairs are also shown, along with the contrasts of the two semi-determinate (S) cultivars *versus* four or six indeterminate (I) cultivars.

[‡] P92M61 and P93M11 in 2006 and P93M61 and P93M11 in 2007 are omitted from 2-yr ANOVA.

prone to lodge when subjected to the wind and rain of summer thunderstorms. The R3 and DF lodging scores were not statistically different from each other, but both were greater than those recorded for the NI treatment. Plants lodged more in 2007 than in 2006, the result of a brief but severe thunderstorm on 8 Aug. 2007 (DAE = 92), which unleashed a torrent of rain accompanied by strong wind gusts that resulted in extensive lodging on that day. The taller plants in the FS treatment lodged more than those in the other treatments—a score of 3.0 denotes that most plants in a plot were leaning 45° from the vertical—and as noted later, this affected yield in this treatment.

Soybean seed yield was enhanced by irrigation relative to the rainfed check (Fig. 2G and 2H). Of great interest (in terms of the research hypothesis stated above) was that the yield ranking for the treatments—FS = R3 > DF > NI—was the same each year. Although the first irrigation date in R3 was deferred for 12 d (2006) and 8 d (2007) after the first irrigation was applied in FS (Fig. 1A and 1B), and a respective 11 and 5 d after R3 were subsequently required to bring the cumulative water supply in these two treatments to equivalence, the yield attained in the R3 treatment each year was *not* significantly diminished relative to the yield attained in the FS treatment. In contrast, the DF treatment, which received only two-thirds of the irrigation amount (single events

and seasonal total) applied in the FS treatment, produced significantly *less* yield than R3 and FS each year—a result that is consistent with the fact that seed yield is linearly related to transpiration (Specht et al., 1986, 2001). The results for IWUE presented in Fig. 2G and H document that although the R3 and FS treatments received the same total seasonal irrigation amount in each year, the greater yield attained in R3 resulted in a higher IWUE than FS. The DF treatment also had a greater IWUE than that for FS, but the R3 and DF IWUE values differed only by 0.4 and 0.2 kg ha⁻¹ per ha-m of water each year, indicating that an R3 strategy instead of a DF strategy does not have much of an impact on IWUE, primarily because the former, despite one-third more irrigation than the latter, produced a compensating greater yield.

Two components of seed yield—plant density (measured at maturity) and seed mass (measured as 100-seed weight)—were evaluated. No attempt was made here to measure seed number per plant, but it is well known that irrigation at R3 or commencing at R3 and continuing thereafter increases the seed number but has little impact on seed mass (Korte et al., 1983b; Kadhemi et al., 1985b). In this study, plant density was not differentially impacted by water regime in either year (the 2-yr experimental mean was 316,000 plants ha⁻¹), indicating that water stress in the rainfed check (NI) or DF irrigation treatments did not affect plant survival (Table 2).

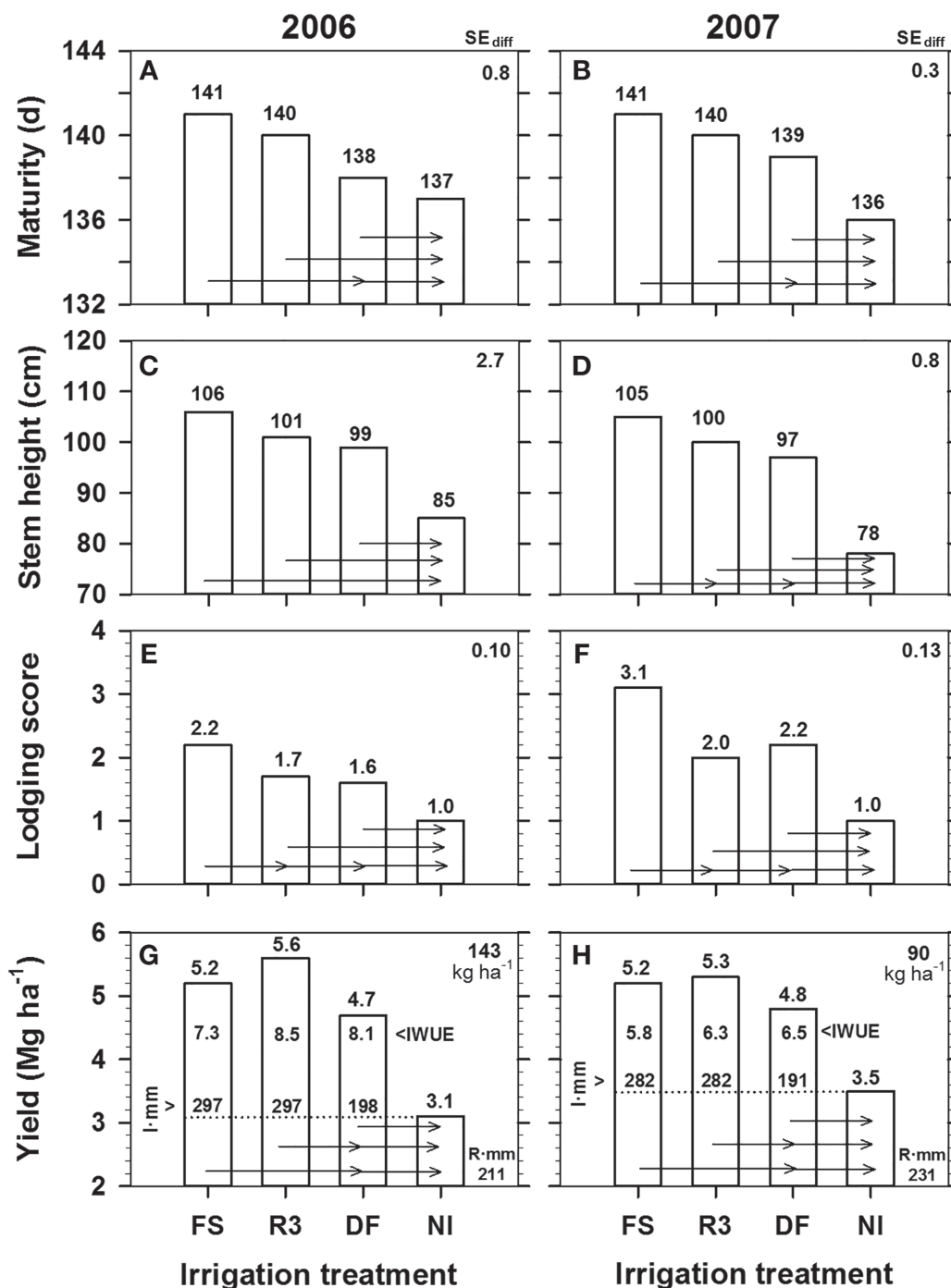


Figure 2. Means of the 100ET-FS, 100ET-R3, 066ET-DF, and 000ET-NI water regimes in 2006 (left) and 2007 (right) for the traits of days to maturity, mature stem height, lodging score, and seed yield. The standard error of the difference (SE_{diff}) between any pair of means (bars) is shown at the upper right of each bar chart of each year and was derived using the Tukey's honest significant difference (HSD) method (adjusted for multiple comparisons). The horizontal arrowed lines denote the significance of the difference between a treatment mean bar in which that line begins and treatment mean bar(s) where an arrowhead ends that line. The dotted line extending leftward from the top of the non-irrigated (NI) yield bar into the full-season (FS), R3, and deficit (DF) yield divides the latter three bars into a yield fraction attributable to total seasonal irrigation amount (I·mm), and a fraction attributable to seasonal rainfall amount (R·mm). Also shown are the $kg\ ha^{-1}\ mm^{-1}$ irrigation water-use efficiency (IWUE) values.

In contrast, seed mass was significantly impacted by water regime in 2006, but not in 2007 (Table 2). The 2006 seed mass means were 16.0 (FS), 16.3 (R3), 15.5 (DF), and 14.8 (NI) $g\ 100\text{-seed}^{-1}$ ($SE_{diff} = 0.1$), with FS and R3 not significantly different from each other, but

with both significantly greater than DF and NI, and with DF significantly greater than NI. The respective means of the seed masses for the 2007 water regime had a narrower (and nonsignificant) range of 18.4 (FS), 18.5 (R3), 18.0 (DF), and 18.8 (NI) $g\ 100\text{-seed}^{-1}$ ($SE_{diff} = 0.3$). The

50% greater August rainfall in 2007 versus that in 2006 (Table 1) resulted in an enhancement of seed mass but also resulted in smaller differences in seed mass between the four water regimes that year.

Relative to the seed constituents of protein and oil (Table 2), the results for the individual years were mixed. In 2006, the means for seed protein content for the four water regimes ranged from 396 g kg⁻¹ (NI) to 402 g kg⁻¹ (R3), and only these two extreme values were significantly different. In 2007, the range was 50% smaller—from 393 g kg⁻¹ (DF) to 396 g kg⁻¹ (FS and R3)—and were not significant. In contrast, the range in seed oil was small and not significant in 2006—from 195 g kg⁻¹ (R3) to 198 g kg⁻¹ (DF)—but in 2007, the range extremes were 225 (NI) and 212 (FS), with seed oil in the NI regime significantly higher than those in all three irrigated treatments. Warm temperatures during the late August through early September seed-fill period are known to enhance soybean seed oil content (Piper and Boote, 1999; Wilson, 2004; Carrera et al., 2009), and September of 2007 was indeed warmer than 2006 (Table 1), which resulted in higher seed oil content in all water regimes in 2007. In addition, it is also known that soybean canopies in irrigated fields tend to be cooler than canopies in rainfed fields because of greater transpirational cooling (Specht et al., 2001). The higher seed oil detected in the 2007 rainfed treatment was thus probably due to warmer canopies in the water-stressed NI plots that year compared to the prior year.

Cultivar Main Effect

The cultivar main effect was statistically significant for all traits in each single-year analysis (Table 2), which was not unexpected given the diversity of the cultivars chosen for this experiment. Preplanned contrasts between the cultivar maturity pairs and between the two types of stem growth habit were conducted to examine in more detail the significant cultivar main effects.

Cultivars of MG 2.6 to 2.7 (i.e., three indeterminates: Asgrow AG2703, Dekalb DKB26-52, and Pioneer P92M61; and one semi-determinate: Nebraska U01-390489) and cultivars of MG 3.0 to 3.1 (i.e., three indeterminates: Asgrow AG3101, Dekalb DKB31-51, and Pioneer P93M11; and one semi-determinate: Nebraska NE3001), were considered to be “adapted” to the test site, though it is known that later-maturing cultivars tend to perform better if not impacted by an early fall frost. In any event, a preplanned single df F-test contrast between the early (E) MG and late (L) MG cultivar paired subsets was conducted for all traits. The E versus L contrast relative to the difference in magnitude and sign (i.e., L – E) was either significant or nonsignificant (ns) for the following traits in the respective years of 2006 (four MG pairs) and 2007 (three MG pairs): maturity (+6.6 and +2.8 d; a 5-d difference was expected), plant stem height (+1.5 [ns] and +2.6 cm), plant lodging score (+0.20 and 0.00

[ns]), plant density (+19,800 [ns] and +24,700 plants ha⁻¹), seed mass (+0.56 and +1.07 g 100-seed⁻¹), seed yield (+495 and +271 kg ha⁻¹), seed protein (+0.56 and +0.39 g kg⁻¹), and seed oil (–0.42 and –0.98 g kg⁻¹). With the exception of seed oil, the MG 3.1 cultivars had greater trait values, although not always significantly so, and, because of later-than-normal fall frosts, yielded more than the four MG 2.6 cultivars, which might lead to the suggestion that MG 3.1 cultivars are technically more “adapted” to the test site. The greater seed-oil content of the MG 2.6 cultivars versus the MG 3.1 cultivars was probably a result of the early-maturing cultivars having seed-fill periods that coincided with warmer late-summer temperatures compared to the seed-fill periods of the later-maturing cultivars (Wilson, 2004).

Another preplanned contrast was a comparison between the early- and late-maturing indeterminate (IN) cultivars and the early- and late-maturing semi-determinate (SD) cultivars (U01-390489, MG 2.7; NE3001, MG 3.0). Some highly productive irrigation systems in NE produce tall plants that can lodge badly, so the shorter-statured SD cultivars may be advantageous in such lodging-prone production systems (Setiyono et al., 2011). The difference in magnitude and sign (i.e., IN – SD) of SD versus IN was significant (or ns) for the following traits in 2006 and 2007: maturity (+2.1 and +4.3 d), plant stem height (+21.1 and +43.1 cm), plant lodging (+0.31 and +1.07), plant density (+92,900 and +91,900 plants ha⁻¹), seed mass (–3.00 and –3.25 g 100-seed⁻¹), seed yield (+87 [ns] and +348 kg ha⁻¹), seed protein (+1.43 and +0.68 g kg⁻¹), and seed oil (–0.38 and –0.10 [ns] g kg⁻¹). With reference to IN cultivars, the two SD cultivars were earlier maturing, shorter, and less lodging prone, but both also had a larger seed size (i.e., greater seed mass). The SD cultivars also produced seeds with significantly less protein but more seed oil (though ns in 2007). The SD and IN cultivars were nearly equivalent in seed yield in 2006 (but not in 2007). The mature plant density of the ID cultivars averaged about 340,000 mature plants ha⁻¹, whereas the large-seed, SD cultivars averaged only about 244,000 mature plants ha⁻¹, primarily because large seeds tend to have more combine-mediated hairline fractures, which makes these seeds more prone to germination failure in cooler, early-spring plantings. Unfortunately, lower germination and the resultant lower final plant density in the large-seed SD cultivars may have resulted in a plant-density-mediated confounding of the observed trait differentials with IN versus SD.

Water Regime × Cultivar Interaction

In a two-factor factorial experiment, the interpretation of main effects is assumed to be meaningless when interaction between factors is significant. However, Hinkelmann and Kempthorne (1994) noted that main effects can be considered interpretable when the interaction is deemed

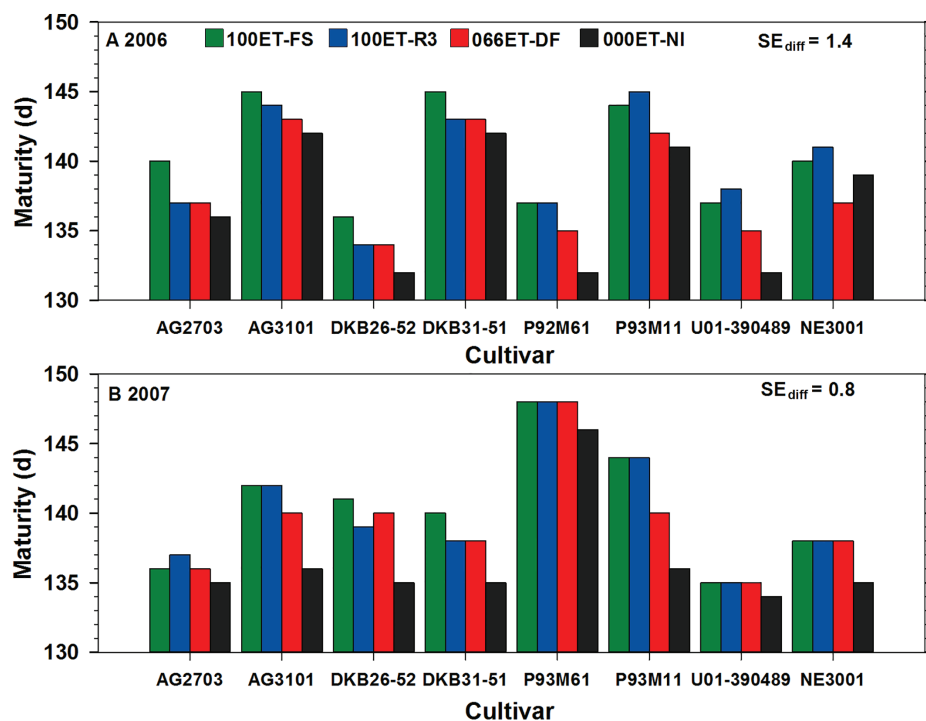


Figure 3. Mean days to maturity for the water regime $\times$ cultivar interaction in 2006 (A) and 2007 (B). The chart bar means for the full-season (FS), R3, deficit (DF), and non-irrigated (NI) water regimes are clustered by cultivar. The Tukey hsd method (adjusted for multiple comparisons) was used to derive a standard error of the difference (SE_{diff}) between *any* pair of interaction means in a given bar chart, or *any* pair of cultivar means *within any* particular irrigation treatment. Note that P93M61 was inadvertently grown in 2007 (instead of the intended P92M61 grown in 2006).

codirectional, but not antidiagonal (i.e., criss-cross), and that is the approach used here. (For implementation details, see Hinkelmann, 2004.)

Except for mature plant density in each year, maturity in 2006, and seed protein in 2007, the interaction between water regime and cultivar was significant for all traits in each year (Table 2). The interactions for the four traits shown in Fig. 2 are graphically depicted in Fig. 3 to 6. The PROC MIXED LSMEANS SLICE option was used to assess $I \times C$ interaction, because it tests the equivalence of LS simple effect means for all levels of one factor within each of the levels of the other factor, and vice versa (SAS Institute, 2008).

The maturity $I \times C$ interaction in 2006 (Fig. 3A) was minimal (ns) but in 2007 (Fig. 3B) was significant (Table 2). The SLICE test (data not shown) indicated that the C-level LS mean simple effects were not equivalent to each other in any of the four I levels in either 2006 or in 2007. The four early/late cultivar pairs displayed the expected alternating pattern of early/late maturity response to I in 2006 (Fig. 3A), but this differential pattern was suppressed in 2007 (Fig. 3B) if one ignores the cultivar P93M61, which was inadvertently planted that year. The observed cultivar maturity responses to the FS, R3, DF, and NI treatments were predominantly codirectional, differing mainly in response magnitude rather than response rank (except for the semi-determinate cultivars in 2006 and the AG2703 and DKB26-52 cultivars in 2007). Thus,

the maturity $I \times C$ interaction did not seem to preclude consideration of the overall FS-R3-DF-NI main effect I pattern depicted in Fig. 2A and 2B.

Stem height $I \times C$ was significant (Table 2) in 2006 (Fig. 4A) and 2007 (Fig. 4B) and was due in part to the semi-determinate cultivars not showing much irrigation-induced stem-height increases compared to the indeterminate cultivars. The SLICE test (data not shown) indicated that the eight C-level simple-effect LS means were not equivalent within any of the four I levels in 2006 or in 2007, nor were the four I-level simple-effect LS means within each of the eight C levels except for NE3001 in 2006 (Fig. 4A). The individual C response patterns in each year (Fig. 4A and 4B) were not appreciably different from the main effect response pattern each year (Fig. 2C and 2D), indicating that $I \times C$ was mostly codirectional. The key anomaly between years was the DF treatment generating smaller stem-height increases relative to NI in 2006 than in 2007. If the goal of using a DF strategy of applying only 66% of the irrigation amount applied in FS and R3 is to keep irrigated plants as short as possible (to lessen lodging risk), then that goal will be difficult to achieve if, in some dry spring years, the first DF irrigation has to be applied before stage R3, because irrigation before that stage will stimulate plant stem elongation.

A significant plant-lodging $I \times C$ interaction (Table 2) was detected each year (Fig. 5) and tended to mimic the

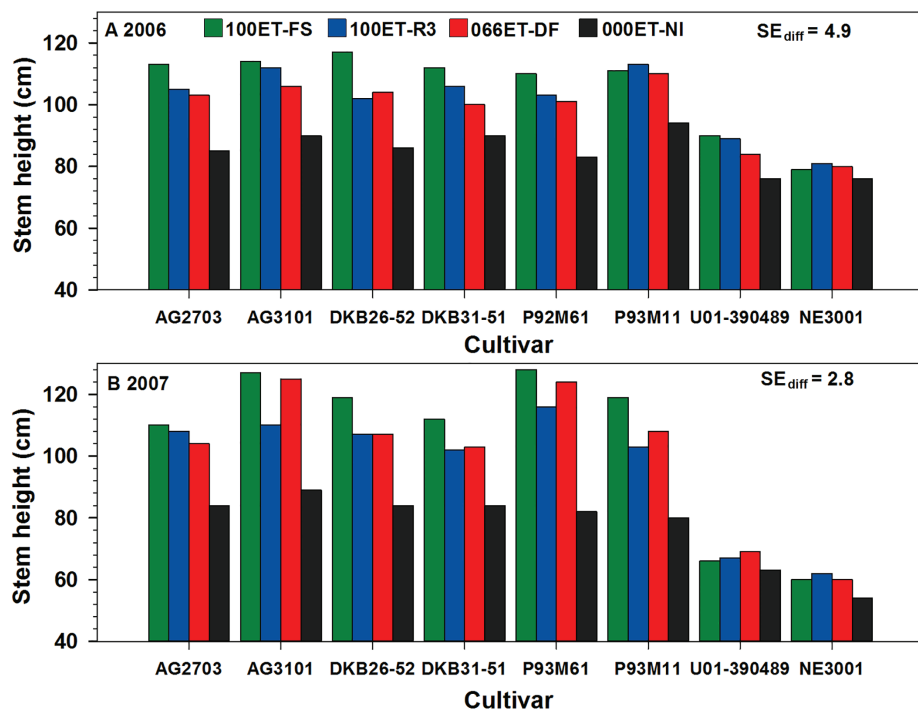


Figure 4. Mean stem height for the water regime $\times$ cultivar interaction in 2006 (A) and 2007 (B). The chart bar means for the full-season (FS), R3, deficit (DF), and non-irrigated (NI) water regimes are clustered by cultivar. The Tukey hsd method (adjusted for multiple comparisons) was used to derive a standard error of the difference (SE_{diff}) between any pair of interaction means in a given bar chart or between any pair of cultivar means within any particular irrigation treatment. Note that P93M61 was inadvertently grown in 2007 (instead of the intended P92M61 grown in 2006).

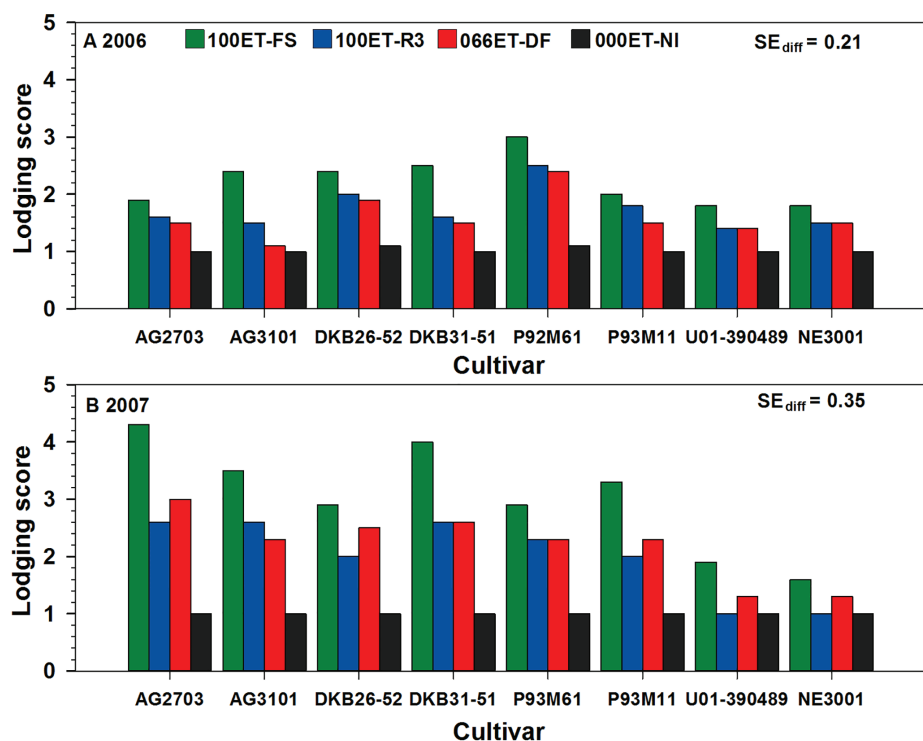


Figure 5. Mean lodging score for the water regime $\times$ cultivar interaction in 2006 (A) and 2007 (B). The chart bar means for the full-season (FS), R3, deficit (DF), and non-irrigated (NI) water regimes are clustered by cultivar. The Tukey hsd method (adjusted for multiple comparisons) was used to derive a standard error of the difference (SE_{diff}) between any pair of interaction means in a given bar chart or between any pair of cultivar means within any particular irrigation treatment. Note that P93M61 was inadvertently grown in 2007 (instead of the intended P92M61 grown in 2006).

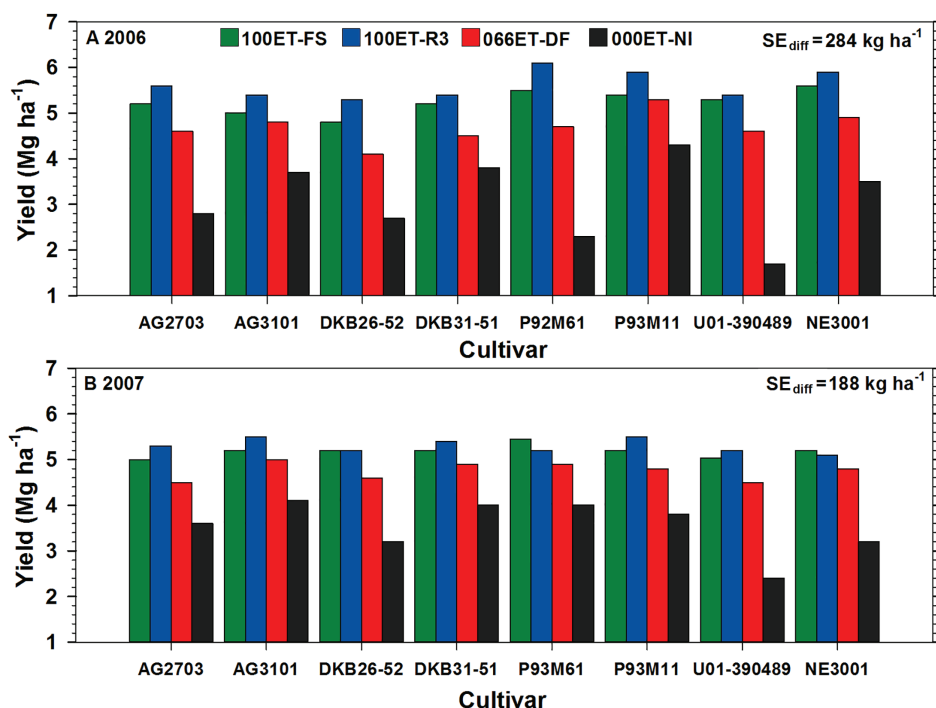


Figure 6. Mean seed yield for the water regime $\times$ cultivar interaction in 2006 (A) and 2007 (B). The chart bar means for the full-season (FS), R3, deficit (DF), and non-irrigated (NI) water regimes are clustered by cultivar. The Tukey hsd method (adjusted for multiple comparisons) was used to derive a standard error of the difference (SE_{diff}) between any pair of interaction means in a given bar chart or between any pair of cultivar means within any particular irrigation treatment. Note that P93M61 was inadvertently grown in 2007 (instead of the intended P92M61 grown in 2006).

height $I \times C$ pattern (Fig. 4), reflecting a commonly observed association between plant tallness and lodging potential. The lodging $I \times C$ each year was mainly due to differences in the cultivar lodging response to irrigation beyond the near-unity NI lodging scores. Compared to 2006 (Fig. 5A), the 2007 FS treatment (Fig. 5B) generated much higher cultivar lodging scores than did DF and R3, and these two generated higher lodging scores than NI. Again, it is clear that a DF strategy, while generating less lodging relative to an FS treatment, did not do so consistently across years relative to lodging generated in the R3 treatment.

The seed yield $I \times C$ interaction (Fig. 6A and 6B) was significant in each year (Table 2). Yet, the yield response of each of the eight cultivars to the four water regimes exhibited a consistent pattern—FS = R3 > DF > NI—in each year, with the R3 treatment yield not significantly less than the FS treatment yield in any cultivar. Likewise, the SLICE test (data not shown) indicated that the simple effects of the C-level LS mean yields in the FS treatment were not significantly different from each other in either year (2006: $P = 0.14$; 2007: $P = 0.37$), as were those in the R3 treatment except for marginal significance in 1 yr (2006: $P = 0.02$; 2007: $P = 0.26$). Note that all eight cultivars produced FS or R3 yields of at least 5 Mg ha⁻¹ each year. In contrast, the C LS mean simple effects in the NI treatment were highly significantly different each year (2006: $P < 0.0001$; 2007: $P < 0.0001$). The left-to-right positional order of the

cultivars in Fig. 6 is coded ABCD[E]FGH (for brevity of discussion) to describe the high-to-low cultivar yield rank in the NI treatment, which was F (P93M11) > D (DKB31-41) $\geq$ B (AG3101) $\geq$ H (NE3001) > A (AG2703) $\geq$ C (DKB26-52) $\geq$ [E] (P92M61) > G (U01-390489) in 2006 (Fig. 6A), and B $\geq$ [E] $\geq$ D > F $\geq$ A > H $\geq$ C > G in 2007 (Fig. 6B). The bracketed E code identifies cultivar P92M61 planted in 2006 and P93M61 planted in error in 2007, and the $\geq$ symbol was used to denote a successive yield rank that was not statistically significant. Omitting P92M61 and P93M61 and considering that the BDFH cultivars were MG 3.0–3.1, whereas the ACG cultivars were MG 2.6–2.7, the reversals in cultivar yield rank for the NI treatment that were of statistical significance were essentially confined to the top three highest-yielding cultivars in the later-maturing class, that is, FDBH in 2006 to BDFH in 2007, although there was also a significant yield rank reversal between A and H in those years. The seasonal pattern of water stress differed between 2006 and 2007 (Fig. 1), with the near-double 2007 August rainfall perhaps contributing to the observed significant changes in yield rank between those years. The lack of stability in cultivar yield ranking under rainfed conditions makes it difficult to recommend a cultivar from among the top-yielding three in the late-maturing class as the best one to use in NI rainfed production systems.

Combined 2-Year Analysis

As was previously noted, data for the cultivars P92M61 (2006) and P93M61 (2007) and P93M11 in both years had to be removed from the 2-yr data analysis (Table 2), and this removal reduced the statistical power because of fewer than expected df for estimation of 2-yr main effects and interactions. However, the significance of the combined data effects of water regime (all traits but plant density and seed protein), cultivar (all traits), and their interaction (all traits but plant density and seed protein) was consistent with the significance detected (for the same stated traits) in the single-year analyses. In any event, the primary interest in the 2-yr analyses was a focus on the interaction terms that included year as a component.

The interaction of year with water regime was not significant for stem height, plant density, or seed protein but was significant for other traits. Those $Y \times I$ interactions were codirectional, except for plant lodging, in which there was a rank reversal between R3 and DF between years (Fig. 2). As one might expect, the year $\times$ cultivar interaction was significant for all traits but seed mass (Table 2), indicating that cultivar seed mass was not impacted by the 2006–2007 differential in August rainfall. Finally, the three-way interaction ($Y \times I \times C$) was statistically significant for maturity, lodging, seed protein, and seed mass (Table 2), though except for seed protein ($P = 0.006$), the degree of three-way interaction significance for the other traits was nominal (P ranged from 0.01 to 0.02). The higher August rainfall in 2007, compared to that in 2006 (Table 1) probably contributed to inconsistency of $I \times C$ interaction for these three traits between the 2 yr. Note that the three-way interaction for seed yield was not significant.

A risk/reward disadvantage of the 100ET-FS strategy is that when an irrigation event is indeed scheduled before R3, unexpected rainfall events could occur just afterward, rendering that irrigation an ill-timed and costly mistake. Such a scenario would not occur in a 100ET-R3 strategy since no irrigation is ever commenced before R3. We estimated that in most years, a producer using a center pivot with a defer-irrigation-to-R3 strategy could potentially avoid applying 0.77, 1.03, or 1.28 ha-cm (i.e., 0.75, 1.00, or 1.25 ac-in) of irrigation water that is typically applied before R3 with an FS strategy—assuming center-pivot irrigation pump capacities of 1325, 1703, or 2082 L h⁻¹ (i.e., 350, 450, or 550 g h⁻¹) (see Table 5 in Martin et al., 2011).

CONCLUSIONS

The same total amount of irrigation water was applied each year to the 100ET-FS and 100ET-R3 treatments. This equivalency was intentional because the comparative focus of interest in this experiment was to determine if forcing a late-April-planted crop to depend solely on stored soil water until the onset of soybean stage R3 would result in plant water stress that might, before R3, reduce seed yield.

However, in this experiment, yield was *not* reduced by deferral of irrigation to R3, despite the paucity of May–June rainfall each year, which fortuitously limited the periodic refilling of soil water storage as the water therein was being depleted by root hairs to meet transpiration demand. Thus, the strategy of deferring all irrigation until the onset of soybean stage R3 is not a *deficit* irrigation strategy, which the 066ET-DF treatment was designed to be. Instead, the underlying promise of a *deferral* irrigation strategy is to avoid irrigation when the yield sensitivity of developmental stage to coincident water stress is low, and instead, allow the crop to more extensively draw on the stored soil water, as shown by Andriani et al. (1991). A caveat about the R3 strategy is that its practical feasibility is limited to deep, medium-to-fine texture soils that are at 100% FC at emergence. Fortunately, the latter is a historical expectation for the eastern half of Nebraska (Sharma and Irmak, 2012a, 2012b) and for most of the Western Corn Belt (Grassini et al., 2010).

Acknowledgments

This soybean research was funded by the Water, Energy, and Agriculture Initiative (WEAI) at the University of Nebraska. WEAI receives its funding from the Nebraska Soybean Board, Nebraska Corn Board, Nebraska Public Power District, and University of Nebraska Agricultural Research Division. We also thank the Nebraska Soybean Board and United Soybean Boards for other indirect funding support of this research.

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